

## Browser (client)

Camera  
(video frames)

MediaPipe Hands  
21 landmarks/frame

Landmark JSON  
(never video)

## Backend

Gates:  $\geq 15$  frames,  
 $\geq 10$  fps, 0.5–10 s

Isotropic coords;  
mirror to user view

Gaussian smoothing  
( $\sigma = 60$  ms)

Resample to  
64 points

Z validity  
(gate only)

Idle trimming  
(palm speed)

13 weighted  
features

Keyed orthonormal  
 $Q_v$  ( $13 \times 13$ )

DTW  $\times$  rotations  
 $\{0, \pm 6, \pm 12\}^\circ$

**Median of 3  
 $\leq 0.216$  (dev.)**

vs  $S_1, S_2, S_3$   
(stored separately)

Gate / Z-validity failure  $\rightarrow$  capture failure (retry), never a mismatch.